

Nathan Liu

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EDUCATION

University of California, Berkeley

May 2026

Bachelor of Arts in Data Science & Computer Science

GPA: 3.84/4.00

- **Coursework:** Structure and Interpretation of Computer Programs, Data Structures, Machine Structures, Introduction to Data Science, Probability for Data Science, Principles & Techniques of Data Science, Introduction to Full-stack Web Development, Efficient Algorithms and Intractable Problems, Optimization Models in Engineering

EXPERIENCE

Operations Data Analyst Intern

June 2024 – Sept. 2024

Happy Lemon USA

Danville, CA

- Designed and implemented a centralized SQL database consolidating data from sales, inventory, and POS, enabling real-time operational tracking.
- Engineered a robust data pipeline using Python (Pandas, mysql.connector) automating the aggregation of 200+ records daily from POS and stock systems.
- Utilized Seaborn to generate critical reports and visualizations, enhancing owners understanding of operational trends across products and sales.
- Analyzed integrated datasets, identifying key trends that contributed to an estimated 15% reduction in inventory spending, equating to roughly \$80,000 in annualized savings.

Happy Lemon USA

Sept. 2021 – Sept. 2024

Shift Manager

Danville, CA

- Led and scheduled a team of 15+ employees, ensuring operational efficiency and productivity over three years.
- Trained and onboarded 10+ new staff members, optimizing both front and back-of-house workflows.
- Managed daily operations, resolved customer concerns, and fostered a positive team environment through effective and consistent communication.

PROJECTS

Predictive YouTube Title Analysis | *Python, Flask, React, GCP, Gemini API*

Fall 2025

- Engineered a full-stack AI application analyzing YouTube title effectiveness (87% accuracy), featuring a Python backend (Flask) hosted on GCP and a React frontend.
- Built the core analysis engine in Python, applying NLP techniques (tokenization, TF-IDF using scikit-learn) and training a Logistic Regression model to classify titles based on engagement potential.
- Leveraged the Gemini API within the Python backend service to generate sophisticated, context-aware performance ratings beyond the primary classification and achieved 94% higher view count correlation.

Personal Portfolio | *JavaScript, HTML/CSS, ImageMagick, Wrangler, S3 API, Cloudflare*

Fall 2025

- Built a full-stack personal website with dynamic APIs and image processing using Express.js, Node.js, and ImageMagick.
- Integrated S3 buckets for efficient media storage and retrieval with dynamic uploading to Cloudflare and automatic website update.
- Deployed via Cloudflare with GitHub CI/CD and Wrangler for automated builds and deployments.

SpotifySync | *JavaScript, HTML/CSS, MongoDB, WebSockets*

Summer 2024

- Developed a real-time collaborative playlist builder (Spotify API) featuring a synchronized swipe UI requiring mutual 'likes' for song addition.
- Engineered the full-stack application using WebSockets for live synchronization, React.js (front-end), Node.js/Express.js (back-end), and MongoDB (session data).
- Implemented CI/CD pipelines via GitHub Actions to automate testing and linting, enhancing workflow efficiency.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, TypeScript, HTML/CSS

Framework & Libraries: Next.js Express.js, PyTorch, Pandas, NumPy, Seaborn, Matplotlib, MongoDB, Django

Developer Tools: Git, Docker, Cloudflare, AWS, GCP, Jupyter Notebook, VS Code